THERMOPAC ENGINEERING LIMITED
Design Software — Engineering Report

ECR Calculation Report (Agitated Column — Stage C5)

Test

Document Number **LLX-RND-2026-0001-ECRR**
Report Revision **Rev 0**
Design Number **LLX-RND-2026-0001**
Design Revision **Rev 0 (draft)**
Module **Liquid-Liquid Extraction**
Design Type **R&D / Independent Design**
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Signature / Date	Signature / Date	Signature / Date

Traceability & Generation Record

This document was generated from a frozen report payload linked to Design Revision Rev 0 (internal id 6) of design LLX-RND-2026-0001. It does not render live database values; regeneration from the same payload reproduces this document exactly.

Calculation runs consumed:

- llx-ecr 1.0.0 — run #64 (2026-08-06 11:25 UTC)

Renders the persisted 'ecr' result snapshot (computed 2026-08-06 11:25 UTC). Applicability: PRELIMINARY ECR-TYPE AGITATED COLUMN SCREENING — NOT VENDOR RATING AND NOT FOR FABRICATION

Generated 2026-08-06 15:07 UTC by Manager.

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1. Introduction, Design Basis & Run Identification

This ECR Calculation Report presents the Stage C5 agitated-column screening for design LLX-RND-2026-0001, Test, rendered from the frozen 'ecr' result snapshot of Design Revision Rev 0. Calculation run: #64 (engine llx-ecr v1.0.0, status 'warning', calculated 2026-08-06 11:25 UTC). The engine was NOT re-run for this report.

Engine chain versions (verbatim from snapshot): cel v1.0.0, epd v1.0.0, processDesign v1.0.0, hydraulicsCommon v1.0.0, ecrAgitatedColumn v1.0.0. Governing basis: LLX-RND-2026-0001-DBR; process flows: LLX-RND-2026-0001-PDR; generic hydraulics: LLX-RND-2026-0001-HDR (same revision).

Engine applicability statement (verbatim): "PRELIMINARY ECR-TYPE AGITATED COLUMN SCREENING — NOT VENDOR RATING AND NOT FOR FABRICATION".

2. Design Basis Inputs

Item	Value	Unit	Source Type	Source Reference
Operating temperature	70	°C	Design basis	—
Phase configuration	nmp_continuous_rrbo_dispersed	—	Engineer Entry	Phase continuity is taken from the engineer input only.
Dispersed / continuous phase	RRBO / NMP	—	Engineer Entry	—
Theoretical stages (from C2)	6	—	C2 frozen result	—
Rotor type	Kühni turbine (default label)	—	Default label	Refinement R2: both entered !Ò ±1 % consistency enforced; one entered !Ò the other calculated
Rotor-to-column diameter ratio D_R/D	0.5	-	Assumed	Thermopac Preliminary Equipment Screening Default v1.0
Compartment height	0.25	m	Assumed	Thermopac Preliminary Equipment Screening Default v1.0
Compartment efficiency	0.4	-	Assumed	Thermopac Preliminary Equipment Screening Default v1.0
Power-density basis	continuous_phase	—	Engine configuration	Refinement R1: selectable basis recorded with every calculation. 'holdup_corrected' is reserved for a future validated-holdup path.
Interfacial tension	0.01	N/m	Assumed	Two-Phase Properties workspace entry (engineer source type: Assumed)
Continuous-phase viscosity	0.00091700000000000001	Pa.s	Assumed — Pending Validation	Assumed: Fluid Properties workspace entry (dynamic viscosity, mPa·s converted to Pa·s)
RRBO density used	870	kg/m3	Entered/EPD	Assumed: Thermopac Feed Master (Default) — Fluid Properties auto-populated value, no engineer source type selected
NMP density used	986.2727272727273	kg/m3	Entered/EPD	Source-tagged tabular data (no fitted correlation approved yet) — CRC Press / provisional (2016)
System derating factor	1	—	Not supplied — default 1.0	Not supplied — 1.0 applied with warning; never invented
Tip-speed criteria	Not supplied	—	Not supplied	Refinement R3: classified below/within/above preferred range and above vendor limit
Vendor hydraulic capacity	Not supplied	—	Not supplied	Not supplied — utilization Pending Validation; never invented
Utilization screening band	40 – 80	%	Configurable criterion	Configurable screening criterion — not a universal engineering rule

3. Equation Reference Index (From Frozen Snapshot)

Every formula/basis string stored in the frozen snapshot, grouped by equation reference and reproduced verbatim. Controlled statements: C5 section of the Correlation & Equation Register; software-verification status tracked in the V&V equation register (all ECR entries currently Partially Verified).

Equation Ref	Formula / basis strings as stored in the frozen snapshot (verbatim; one line per distinct usage)
ECR-001	A = pi·D²/4 Total both-phase liquid load Q_RRBO/A Q_NMP/A u_d = Q_RRBO/A u_c = Q_NMP/A … 1 further distinct usage string(s) omitted for legibility (they differ only in per-diameter numeric values) — the COMPLETE verbatim set is in the frozen JSON snapshot export accompanying this report.
ECR-002	Vendor Hydraulic Capacity
ECR-003	(Q_c+Q_d)/A_R, A_R = pi·D_R²/4 = 0.01767 m² Stator open-area fraction Vendor stator-velocity limits (Q_c+Q_d)/A_R, A_R = pi·D_R²/4 = 0.02405 m² (Q_c+Q_d)/A_R, A_R = pi·D_R²/4 = 0.03142 m² (Q_c+Q_d)/A_R, A_R = pi·D_R²/4 = 0.03976 m² … 31 further distinct usage string(s) omitted for legibility (they differ only in per-diameter numeric values) — the COMPLETE verbatim set is in the frozen JSON snapshot export accompanying this report.
ECR-004	Assumed: Thermopac Preliminary Equipment Screening Default v1.0 N = rpm/60 A_R = pi·D_R²/4 Calculated from ratio 0.5 (Assumed: Thermopac Preliminary Equipment Screening Default v1.0) × column diameter
ECR-005	We = rho_c·N²·D_R³/sigma (sigma 0.01 N/m [Assumed: Two-Phase Properties workspace entry (engineer source type: Assumed)]) Fr = N²·D_R/g Re = rho_c·N·D_R²/μ_c (rho_c 986.3 kg/m³ [Source-tagged tabular data (no fitted correlation approved yet) — CRC Press / provisional (2016)], μ_c 0.0009170000000000001 Pa·s [Assumed: Fluid Properties workspace entry (dynamic viscosity, mPa·s converted to Pa·s)])
ECR-006	P_1 = N_P 1 [Assumed: Thermopac Preliminary Equipment Screening Default v1.0] · rho_m 986.3 kg/m³ [basis 'continuous_phase'] · N³ · D_R^5 P_1 [rho_m 986.3 kg/m³, basis 'continuous_phase'] × 15 rotors (15 compartments × 1 rotor(s)/compartment) P_shaft [rho_m 986.3 kg/m³, basis 'continuous_phase'] / eta_shaft 0.9 [Assumed: Thermopac Preliminary Equipment Screening Default v1.0] × margin 1.25 [Assumed: Thermopac Preliminary Equipment Screening Default v1.0]
ECR-007	ceil(theoreticalStages 6 / compartmentEfficiency 0.4 [Assumed: Thermopac Preliminary Equipment Screening Default v1.0])
ECR-008	Assumed: Thermopac Preliminary Equipment Screening Default v1.0 ECR-007/ECR-008: 15 compartments × 0.25 m (Assumed: Thermopac Preliminary Equipment Screening Default v1.0) Total T/T + top head + bottom head + drive/seal/bearing allowance 15 compartments × compartment height Sum of tangent-to-tangent items (heads and drive/seal excluded)
ECR-009	v_tip = pi·D_R·N pi·D_R·N vs preferred range and vendor limit maxAllowableShaftPower No tip-speed criteria supplied maxUnsupportedShaftLength Vendor/mechanical data

4. Compartments & Active Height

Item	Value	Unit	Source Type	Source Reference
Number of compartments [ECR-007]	15	-	Pending Validation	ceil(theoreticalStages 6 / compartmentEfficiency 0.4 [Assumed: Thermopac Preliminary Equipment Screening Default v1.0])
Active agitated height [ECR-008]	3.75	m	Pending Validation	15 compartments × compartment height

5. Case Flows — Normal

Item	Value	Unit	Source Type	Source Reference
NMP (continuous) mass flow	5917.64	kg/h	From C2 frozen flows	—
RRBO (dispersed) mass flow	3480.00	kg/h	From C2 frozen flows	—
NMP volumetric flow	6.00	m³/h	From C2 frozen flows	—
RRBO volumetric flow	4.00	m³/h	From C2 frozen flows	—

6. Rotor Geometry — Normal Case

D_R from D_R/D ratio (ECR-004); swept area A_R = pi·D_R²/4 (ECR-003/ECR-005 basis strings stored per diameter).

D (m)	D_R (m)	D_R/D (–)	Swept area A_R (m²)	Rotor type
0.30	0.150	0.50	0.0177	Kühni turbine (default label)
0.35	0.175	0.50	0.0241	Kühni turbine (default label)
0.40	0.200	0.50	0.0314	Kühni turbine (default label)
0.45	0.225	0.50	0.0398	Kühni turbine (default label)
0.50	0.250	0.50	0.0491	Kühni turbine (default label)
0.55	0.275	0.50	0.0594	Kühni turbine (default label)
0.60	0.300	0.50	0.0707	Kühni turbine (default label)
0.65	0.325	0.50	0.0830	Kühni turbine (default label)
0.70	0.350	0.50	0.0962	Kühni turbine (default label)
0.75	0.375	0.50	0.1104	Kühni turbine (default label)
0.80	0.400	0.50	0.1257	Kühni turbine (default label)
0.85	0.425	0.50	0.1419	Kühni turbine (default label)
0.90	0.450	0.50	0.1590	Kühni turbine (default label)
0.95	0.475	0.50	0.1772	Kühni turbine (default label)
1.00	0.500	0.50	0.1963	Kühni turbine (default label)
1.05	0.525	0.50	0.2165	Kühni turbine (default label)
1.10	0.550	0.50	0.2376	Kühni turbine (default label)
1.15	0.575	0.50	0.2597	Kühni turbine (default label)

D (m)	D_R (m)	D_R/D (–)	Swept area A_R (m²)	Rotor type
1.20	0.600	0.50	0.2827	Kühni turbine (default label)
1.25	0.625	0.50	0.3068	Kühni turbine (default label)
1.30	0.650	0.50	0.3318	Kühni turbine (default label)
1.35	0.675	0.50	0.3578	Kühni turbine (default label)
1.40	0.700	0.50	0.3848	Kühni turbine (default label)
1.45	0.725	0.50	0.4128	Kühni turbine (default label)
1.50	0.750	0.50	0.4418	Kühni turbine (default label)
1.55	0.775	0.50	0.4717	Kühni turbine (default label)
1.60	0.800	0.50	0.5027	Kühni turbine (default label)
1.65	0.825	0.50	0.5346	Kühni turbine (default label)
1.70	0.850	0.50	0.5675	Kühni turbine (default label)
1.75	0.875	0.50	0.6013	Kühni turbine (default label)
1.80	0.900	0.50	0.6362	Kühni turbine (default label)
1.85	0.925	0.50	0.6720	Kühni turbine (default label)
1.90	0.950	0.50	0.7088	Kühni turbine (default label)
1.95	0.975	0.50	0.7466	Kühni turbine (default label)
2.00	1.000	0.50	0.7854	Kühni turbine (default label)

7. Rotor Speed, Tip Speed & Dimensionless Groups — Normal Case

Single speed point per diameter (N = 60 rpm, Assumed screening default). Re = rho_c·N·D_R²/μ_c; We = rho_c·N²·D_R³/sigma; Fr = N²·D_R/g; v_tip = pi·D_R·N (ECR-005/ECR-009). No droplet-size prediction is made from We (engine statement, verbatim).

D (m)	N (rpm)	v_tip (m/s)	Re (–)	We (–)	Fr (–)	Tip-speed class
0.30	60	0.471	24200	332.9	0.0153	no_limit_data
0.35	60	0.550	32938	528.6	0.0178	no_limit_data
0.40	60	0.628	43022	789.0	0.0204	no_limit_data
0.45	60	0.707	54449	1123.4	0.0229	no_limit_data
0.50	60	0.785	67221	1541.1	0.0255	no_limit_data
0.55	60	0.864	81338	2051.1	0.0280	no_limit_data
0.60	60	0.942	96799	2662.9	0.0306	no_limit_data
0.65	60	1.021	113604	3385.7	0.0331	no_limit_data
0.70	60	1.100	131754	4228.6	0.0357	no_limit_data
0.75	60	1.178	151248	5201.0	0.0382	no_limit_data
0.80	60	1.257	172087	6312.1	0.0408	no_limit_data
0.85	60	1.335	194270	7571.2	0.0433	no_limit_data

D (m)	N (rpm)	v_tip (m/s)	Re (–)	We (–)	Fr (–)	Tip-speed class
0.90	60	1.414	217797	8987.4	0.0459	no_limit_data
0.95	60	1.492	242669	10570.1	0.0484	no_limit_data
1.00	60	1.571	268886	12328.4	0.0510	no_limit_data
1.05	60	1.649	296446	14271.7	0.0535	no_limit_data
1.10	60	1.728	325352	16409.1	0.0561	no_limit_data
1.15	60	1.806	355601	18750.0	0.0586	no_limit_data
1.20	60	1.885	387195	21303.5	0.0612	no_limit_data
1.25	60	1.963	420134	24078.9	0.0637	no_limit_data
1.30	60	2.042	454417	27085.5	0.0663	no_limit_data
1.35	60	2.121	490044	30332.5	0.0688	no_limit_data
1.40	60	2.199	527016	33829.2	0.0714	no_limit_data
1.45	60	2.278	565332	37584.7	0.0739	no_limit_data
1.50	60	2.356	604993	41608.4	0.0765	no_limit_data
1.55	60	2.435	645998	45909.5	0.0790	no_limit_data
1.60	60	2.513	688347	50497.2	0.0816	no_limit_data
1.65	60	2.592	732041	55380.8	0.0841	no_limit_data
1.70	60	2.670	777080	60569.5	0.0867	no_limit_data
1.75	60	2.749	823462	66072.6	0.0892	no_limit_data
1.80	60	2.827	871190	71899.3	0.0918	no_limit_data
1.85	60	2.906	920261	78058.9	0.0943	no_limit_data
1.90	60	2.985	970677	84560.6	0.0969	no_limit_data
1.95	60	3.063	1022438	91413.6	0.0994	no_limit_data
2.00	60	3.142	1075543	98627.3	0.1020	no_limit_data

8. Agitation & Motor Power — Normal Case

P_1 = N_P·rho_m·N³·D_R^5 per rotor (N_P = 1 Assumed); P_shaft = P_1 × rotors; P_motor = P_shaft/eta_shaft(0.9, Assumed) × margin 1.25 (Assumed) — screening only, not a motor selection (ECR-006).

D (m)	P per rotor (W)	P total shaft (W)	P motor design (W)	Density basis
0.30	0.075	1.123	1.560	continuous_phase
0.35	0.162	2.428	3.372	continuous_phase
0.40	0.316	4.734	6.575	continuous_phase
0.45	0.569	8.531	11.849	continuous_phase
0.50	0.963	14.447	20.066	continuous_phase
0.55	1.551	23.268	32.316	continuous_phase
0.60	2.397	35.950	49.930	continuous_phase
0.65	3.576	53.642	74.503	continuous_phase
0.70	5.180	77.701	107.919	continuous_phase
0.75	7.314	109.710	152.374	continuous_phase
0.80	10.099	151.491	210.405	continuous_phase
0.85	13.675	205.132	284.905	continuous_phase
0.90	18.200	272.993	379.156	continuous_phase
0.95	23.849	357.731	496.848	continuous_phase
1.00	30.821	462.315	642.105	continuous_phase
1.05	39.336	590.045	819.506	continuous_phase
1.10	49.638	744.563	1034.116	continuous_phase

D (m)	P per rotor (W)	P total shaft (W)	P motor design (W)	Density basis
1.15	61.992	929.881	1291.502	continuous_phase
1.20	76.693	1150.389	1597.762	continuous_phase
1.25	94.058	1410.874	1959.548	continuous_phase
1.30	114.436	1716.544	2384.090	continuous_phase
1.35	138.202	2073.037	2879.219	continuous_phase
1.40	165.763	2486.443	3453.393	continuous_phase
1.45	197.555	2963.318	4115.720	continuous_phase
1.50	234.047	3510.707	4875.982	continuous_phase
1.55	275.744	4136.155	5744.660	continuous_phase
1.60	323.182	4847.728	6732.955	continuous_phase
1.65	376.935	5654.029	7852.818	continuous_phase
1.70	437.614	6564.217	9116.968	continuous_phase
1.75	505.868	7588.021	10538.919	continuous_phase
1.80	582.384	8735.763	12133.004	continuous_phase
1.85	667.891	10018.367	13914.399	continuous_phase
1.90	763.159	11447.386	15899.147	continuous_phase
1.95	869.001	13035.010	18104.180	continuous_phase
2.00	986.273	14794.091	20547.348	continuous_phase

9. Hydraulic Loading & Utilization — Normal Case

"Not Calculable" cells reproduce the engine finding — no value was invented. Utilization is not screenable without ECR-specific Vendor Hydraulic Capacity (ECR-002).

D (m)	u_c NMP ($m^3/m^2\cdot h$)	u_d RRBO ($m^3/m^2\cdot h$)	Total ($m^3/m^2\cdot h$)	Swept-area loading ($m^3/m^2\cdot h$)	Stator free-area vel.	Utilization (%)	Feasibility
0.30	84.88	56.59	141.47	565.9	Not Calculable	Pending Validation	pending_validation
0.35	62.36	41.58	103.94	415.8	Not Calculable	Pending Validation	pending_validation
0.40	47.75	31.83	79.58	318.3	Not Calculable	Pending Validation	pending_validation
0.45	37.73	25.15	62.88	251.5	Not Calculable	Pending Validation	pending_validation
0.50	30.56	20.37	50.93	203.7	Not Calculable	Pending Validation	pending_validation
0.55	25.25	16.84	42.09	168.4	Not Calculable	Pending Validation	pending_validation
0.60	21.22	14.15	35.37	141.5	Not Calculable	Pending Validation	pending_validation
0.65	18.08	12.05	30.14	120.5	Not Calculable	Pending Validation	pending_validation
0.70	15.59	10.39	25.98	103.9	Not Calculable	Pending Validation	pending_validation
0.75	13.58	9.05	22.64	90.5	Not Calculable	Pending Validation	pending_validation
0.80	11.94	7.96	19.89	79.6	Not Calculable	Pending Validation	pending_validation
0.85	10.57	7.05	17.62	70.5	Not Calculable	Pending Validation	pending_validation
0.90	9.43	6.29	15.72	62.9	Not Calculable	Pending Validation	pending_validation

D (m)	u_c NMP (m³/m²·h)	u_d RRBO (m³/m²·h)	Total (m³/m²·h)	Swept-area loading (m³/m²·h)	Stator free-area vel.	Utilization (%)	Feasibility
0.95	8.46	5.64	14.11	56.4	Not Calculable	Pending Validation	pending_validation
1.00	7.64	5.09	12.73	50.9	Not Calculable	Pending Validation	pending_validation
1.05	6.93	4.62	11.55	46.2	Not Calculable	Pending Validation	pending_validation
1.10	6.31	4.21	10.52	42.1	Not Calculable	Pending Validation	pending_validation
1.15	5.78	3.85	9.63	38.5	Not Calculable	Pending Validation	pending_validation
1.20	5.31	3.54	8.84	35.4	Not Calculable	Pending Validation	pending_validation
1.25	4.89	3.26	8.15	32.6	Not Calculable	Pending Validation	pending_validation
1.30	4.52	3.01	7.53	30.1	Not Calculable	Pending Validation	pending_validation
1.35	4.19	2.79	6.99	27.9	Not Calculable	Pending Validation	pending_validation
1.40	3.90	2.60	6.50	26.0	Not Calculable	Pending Validation	pending_validation
1.45	3.63	2.42	6.06	24.2	Not Calculable	Pending Validation	pending_validation
1.50	3.40	2.26	5.66	22.6	Not Calculable	Pending Validation	pending_validation
1.55	3.18	2.12	5.30	21.2	Not Calculable	Pending Validation	pending_validation
1.60	2.98	1.99	4.97	19.9	Not Calculable	Pending Validation	pending_validation
1.65	2.81	1.87	4.68	18.7	Not Calculable	Pending Validation	pending_validation
1.70	2.64	1.76	4.41	17.6	Not Calculable	Pending Validation	pending_validation
1.75	2.49	1.66	4.16	16.6	Not Calculable	Pending Validation	pending_validation
1.80	2.36	1.57	3.93	15.7	Not Calculable	Pending Validation	pending_validation
1.85	2.23	1.49	3.72	14.9	Not Calculable	Pending Validation	pending_validation
1.90	2.12	1.41	3.53	14.1	Not Calculable	Pending Validation	pending_validation
1.95	2.01	1.34	3.35	13.4	Not Calculable	Pending Validation	pending_validation
2.00	1.91	1.27	3.18	12.7	Not Calculable	Pending Validation	pending_validation

10. Case Flows — Maximum Continuous

Item	Value	Unit	Source Type	Source Reference
NMP (continuous) mass flow	7101.16	kg/h	From C2 frozen flows	—
RRBO (dispersed) mass flow	3480.00	kg/h	From C2 frozen flows	—
NMP volumetric flow	7.20	m³/h	From C2 frozen flows	—
RRBO volumetric flow	4.00	m³/h	From C2 frozen flows	—

11. Rotor Speed, Tip Speed & Dimensionless Groups — Maximum Continuous Case

D (m)	N (rpm)	v_tip (m/s)	Re (–)	We (–)	Fr (–)	Tip-speed class
0.30	60	0.471	24200	332.9	0.0153	no_limit_data
0.35	60	0.550	32938	528.6	0.0178	no_limit_data
0.40	60	0.628	43022	789.0	0.0204	no_limit_data
0.45	60	0.707	54449	1123.4	0.0229	no_limit_data
0.50	60	0.785	67221	1541.1	0.0255	no_limit_data
0.55	60	0.864	81338	2051.1	0.0280	no_limit_data
0.60	60	0.942	96799	2662.9	0.0306	no_limit_data
0.65	60	1.021	113604	3385.7	0.0331	no_limit_data
0.70	60	1.100	131754	4228.6	0.0357	no_limit_data
0.75	60	1.178	151248	5201.0	0.0382	no_limit_data
0.80	60	1.257	172087	6312.1	0.0408	no_limit_data
0.85	60	1.335	194270	7571.2	0.0433	no_limit_data
0.90	60	1.414	217797	8987.4	0.0459	no_limit_data
0.95	60	1.492	242669	10570.1	0.0484	no_limit_data
1.00	60	1.571	268886	12328.4	0.0510	no_limit_data
1.05	60	1.649	296446	14271.7	0.0535	no_limit_data
1.10	60	1.728	325352	16409.1	0.0561	no_limit_data
1.15	60	1.806	355601	18750.0	0.0586	no_limit_data
1.20	60	1.885	387195	21303.5	0.0612	no_limit_data
1.25	60	1.963	420134	24078.9	0.0637	no_limit_data
1.30	60	2.042	454417	27085.5	0.0663	no_limit_data
1.35	60	2.121	490044	30332.5	0.0688	no_limit_data
1.40	60	2.199	527016	33829.2	0.0714	no_limit_data
1.45	60	2.278	565332	37584.7	0.0739	no_limit_data
1.50	60	2.356	604993	41608.4	0.0765	no_limit_data
1.55	60	2.435	645998	45909.5	0.0790	no_limit_data
1.60	60	2.513	688347	50497.2	0.0816	no_limit_data
1.65	60	2.592	732041	55380.8	0.0841	no_limit_data
1.70	60	2.670	777080	60569.5	0.0867	no_limit_data
1.75	60	2.749	823462	66072.6	0.0892	no_limit_data
1.80	60	2.827	871190	71899.3	0.0918	no_limit_data
1.85	60	2.906	920261	78058.9	0.0943	no_limit_data
1.90	60	2.985	970677	84560.6	0.0969	no_limit_data
1.95	60	3.063	1022438	91413.6	0.0994	no_limit_data
2.00	60	3.142	1075543	98627.3	0.1020	no_limit_data

12. Agitation & Motor Power — Maximum Continuous Case

D (m)	P per rotor (W)	P total shaft (W)	P motor design (W)	Density basis
0.30	0.075	1.123	1.560	continuous_phase
0.35	0.162	2.428	3.372	continuous_phase
0.40	0.316	4.734	6.575	continuous_phase
0.45	0.569	8.531	11.849	continuous_phase
0.50	0.963	14.447	20.066	continuous_phase

D (m)	P per rotor (W)	P total shaft (W)	P motor design (W)	Density basis
0.55	1.551	23.268	32.316	continuous_phase
0.60	2.397	35.950	49.930	continuous_phase
0.65	3.576	53.642	74.503	continuous_phase
0.70	5.180	77.701	107.919	continuous_phase
0.75	7.314	109.710	152.374	continuous_phase
0.80	10.099	151.491	210.405	continuous_phase
0.85	13.675	205.132	284.905	continuous_phase
0.90	18.200	272.993	379.156	continuous_phase
0.95	23.849	357.731	496.848	continuous_phase
1.00	30.821	462.315	642.105	continuous_phase
1.05	39.336	590.045	819.506	continuous_phase
1.10	49.638	744.563	1034.116	continuous_phase
1.15	61.992	929.881	1291.502	continuous_phase
1.20	76.693	1150.389	1597.762	continuous_phase
1.25	94.058	1410.874	1959.548	continuous_phase
1.30	114.436	1716.544	2384.090	continuous_phase
1.35	138.202	2073.037	2879.219	continuous_phase
1.40	165.763	2486.443	3453.393	continuous_phase
1.45	197.555	2963.318	4115.720	continuous_phase
1.50	234.047	3510.707	4875.982	continuous_phase
1.55	275.744	4136.155	5744.660	continuous_phase
1.60	323.182	4847.728	6732.955	continuous_phase
1.65	376.935	5654.029	7852.818	continuous_phase
1.70	437.614	6564.217	9116.968	continuous_phase
1.75	505.868	7588.021	10538.919	continuous_phase
1.80	582.384	8735.763	12133.004	continuous_phase
1.85	667.891	10018.367	13914.399	continuous_phase
1.90	763.159	11447.386	15899.147	continuous_phase
1.95	869.001	13035.010	18104.180	continuous_phase
2.00	986.273	14794.091	20547.348	continuous_phase

13. Hydraulic Loading & Utilization — Maximum Continuous Case

D (m)	u_c NMP ($m^3/m^2 \cdot h$)	u_d RRBO ($m^3/m^2 \cdot h$)	Total ($m^3/m^2 \cdot h$)	Swept-area loading ($m^3/m^2 \cdot h$)	Stator free-area vel.	Utilization (%)	Feasibility
0.30	101.86	56.59	158.45	633.8	Not Calculable	Pending Validation	pending_validation
0.35	74.84	41.58	116.41	465.6	Not Calculable	Pending Validation	pending_validation
0.40	57.30	31.83	89.13	356.5	Not Calculable	Pending Validation	pending_validation
0.45	45.27	25.15	70.42	281.7	Not Calculable	Pending Validation	pending_validation
0.50	36.67	20.37	57.04	228.2	Not Calculable	Pending Validation	pending_validation
0.55	30.31	16.84	47.14	188.6	Not Calculable	Pending Validation	pending_validation

D (m)	u_c NMP (m³/m²·h)	u_d RRBO (m³/m²·h)	Total (m³/m²·h)	Swept-area loading (m³/m²·h)	Stator free-area vel.	Utilization (%)	Feasibility
0.60	25.46	14.15	39.61	158.4	Not Calculable	Pending Validation	pending_validation
0.65	21.70	12.05	33.75	135.0	Not Calculable	Pending Validation	pending_validation
0.70	18.71	10.39	29.10	116.4	Not Calculable	Pending Validation	pending_validation
0.75	16.30	9.05	25.35	101.4	Not Calculable	Pending Validation	pending_validation
0.80	14.32	7.96	22.28	89.1	Not Calculable	Pending Validation	pending_validation
0.85	12.69	7.05	19.74	78.9	Not Calculable	Pending Validation	pending_validation
0.90	11.32	6.29	17.61	70.4	Not Calculable	Pending Validation	pending_validation
0.95	10.16	5.64	15.80	63.2	Not Calculable	Pending Validation	pending_validation
1.00	9.17	5.09	14.26	57.0	Not Calculable	Pending Validation	pending_validation
1.05	8.32	4.62	12.93	51.7	Not Calculable	Pending Validation	pending_validation
1.10	7.58	4.21	11.79	47.1	Not Calculable	Pending Validation	pending_validation
1.15	6.93	3.85	10.78	43.1	Not Calculable	Pending Validation	pending_validation
1.20	6.37	3.54	9.90	39.6	Not Calculable	Pending Validation	pending_validation
1.25	5.87	3.26	9.13	36.5	Not Calculable	Pending Validation	pending_validation
1.30	5.42	3.01	8.44	33.8	Not Calculable	Pending Validation	pending_validation
1.35	5.03	2.79	7.82	31.3	Not Calculable	Pending Validation	pending_validation
1.40	4.68	2.60	7.28	29.1	Not Calculable	Pending Validation	pending_validation
1.45	4.36	2.42	6.78	27.1	Not Calculable	Pending Validation	pending_validation
1.50	4.07	2.26	6.34	25.4	Not Calculable	Pending Validation	pending_validation
1.55	3.82	2.12	5.94	23.7	Not Calculable	Pending Validation	pending_validation
1.60	3.58	1.99	5.57	22.3	Not Calculable	Pending Validation	pending_validation
1.65	3.37	1.87	5.24	21.0	Not Calculable	Pending Validation	pending_validation
1.70	3.17	1.76	4.93	19.7	Not Calculable	Pending Validation	pending_validation
1.75	2.99	1.66	4.66	18.6	Not Calculable	Pending Validation	pending_validation
1.80	2.83	1.57	4.40	17.6	Not Calculable	Pending Validation	pending_validation
1.85	2.68	1.49	4.17	16.7	Not Calculable	Pending Validation	pending_validation
1.90	2.54	1.41	3.95	15.8	Not Calculable	Pending Validation	pending_validation

D (m)	u_c NMP (m³/m²·h)	u_d RRBO (m³/m²·h)	Total (m³/m²·h)	Swept-area loading (m³/m²·h)	Stator free-area vel.	Utilization (%)	Feasibility
1.95	2.41	1.34	3.75	15.0	Not Calculable	Pending Validation	pending_validation
2.00	2.29	1.27	3.57	14.3	Not Calculable	Pending Validation	pending_validation

14. Mechanical Screening — Shaft, Bearings & Tip Speed

Rendered from the first-diameter entries; the stored explanation strings are identical across diameters because no vendor/mechanical data were supplied.

Item	Value	Unit	Source Type	Source Reference
Tip-speed check [ECR-009]	no_limit_data	-	Not Calculable	pi·D_R·N vs preferred range and vendor limit
Shaft-power check [ECR-009]	Not Calculable	-	Not Calculable	maxAllowableShaftPower
Shaft-support check [ECR-009]	Not Calculable	-	Not Calculable	maxUnsupportedShaftLength
Bearing/support requirements [ECR-009]	pending_validation	-	Pending Validation	Vendor/mechanical data
Stator velocity limit check [ECR-003]	Not Calculable	-	Not Calculable	Vendor stator-velocity limits

15. Height Breakdown — Disengagement, T/T & Overall

Internals stack from the frozen snapshot (ECR-008). Disengagement allowances are the Top/Bottom Disengagement lines; the drive/seal/bearing allowance is excluded from T/T and included in the overall height.

Item	Value	Unit	Source Type	Source Reference
Drive/Seal/Bearing [ECR-008]	1.00	m	Pending Validation	Assumed: Thermopac Preliminary Equipment Screening Default v1.0
Top Head [ECR-008]	0.50	m	Pending Validation	Assumed: Thermopac Preliminary Equipment Screening Default v1.0
Top Disengagement [ECR-008]	1.00	m	Pending Validation	Assumed: Thermopac Preliminary Equipment Screening Default v1.0
Top Distributor [ECR-008]	0.50	m	Pending Validation	Assumed: Thermopac Preliminary Equipment Screening Default v1.0
Active Agitated Section [ECR-008]	3.75	m	Pending Validation	ECR-007/ECR-008: 15 compartments × 0.25 m (Assumed: Thermopac Preliminary Equipment Screening Default v1.0)
Bottom Distributor [ECR-008]	0.50	m	Pending Validation	Assumed: Thermopac Preliminary Equipment Screening Default v1.0
Bottom Disengagement [ECR-008]	1.00	m	Pending Validation	Assumed: Thermopac Preliminary Equipment Screening Default v1.0
Bottom Head [ECR-008]	0.50	m	Pending Validation	Assumed: Thermopac Preliminary Equipment Screening Default v1.0
Active agitated height [ECR-008]	3.75	m	Pending Validation	15 compartments × compartment height
Total tangent-to-tangent (T/T) [ECR-008]	6.75	m	Pending Validation	Sum of tangent-to-tangent items (heads and drive/seal excluded)
Overall vessel height [ECR-008]	8.75	m	Pending Validation	Total T/T + top head + bottom head + drive/seal/bearing allowance

16. Technology Comparison — ECP vs ECR (Report-Layer Derivation)

This side-by-side table is derived MECHANICALLY in the report layer from the two frozen C4/C5 snapshots of the same design revision. It is not an engine result and it expresses no technology preference — both technologies remain Pending Validation and neither utilization band can be screened without vendor capacity data.

Item	ECP (Packed Column, C4)	ECR (Agitated Column, C5)
Theoretical stages (from C2)	6	6
Stage device	HETS 0.75 m (Assumed)	15 compartments × 0.25 m (efficiency 0.4, Assumed)
Active height	4.50 m packed (Pending Validation)	3.75 m agitated (Pending Validation)
Total tangent-to-tangent	7.90 m	6.75 m
Overall vessel height	8.90 m	8.75 m
Internals basis	Packing record: SMVP — Preliminary Screening Record (Rev 0 (preliminary screening)) — Assumed screening data, not vendor-certified	Rotor: Kühni turbine (default label); D_R/D ratio 0.5 (Assumed)
Mechanical drive	None (static internals)	Agitated — motor design power is diameter- and speed-dependent (see ECR power tables); screening only, not a motor selection
Hydraulic utilization screening	Not screenable — no Vendor Packing Capacity data in the packing record	Not screenable — no ECR-specific Vendor Hydraulic Capacity data
Pressure drop	Not Calculable — no wet pressure-drop basis in the packing record	Not emitted by the C5 screening scope
Run status	pending_validation	pending_validation

17. Engine Warnings (Verbatim)

No warning strings stored on individual items. Run status: pending_validation — see Validation & Missing-Data Summary.

18. Limitations (Verbatim)

1. No proprietary Kühni model
2. No vendor hydraulic guarantee
3. No validated droplet breakup/coalescence model
4. No axial back-mixing model
5. No rate-based mass-transfer model
6. No final shaft/seal/bearing design
7. No mechanical code design

19. Independent-Verifiability Statement

Items in the table below are NOT independently verifiable from this report: the engine declared them Not Calculable because the governing vendor/laboratory datum is absent, and no value was invented. A checker cannot reproduce a number that was never computed — closure of these items requires the named data, not re-calculation.

In addition, every ECR result carrying status 'Pending Validation' is ARITHMETICALLY verifiable from the inputs and formula strings in this report, but NOT engineering-verifiable: its basis contains Assumed data (see the Assumptions Register). Hand-checking confirms the arithmetic only; it does not validate the assumed values.

Equation Ref	Item (stored basis string)	Why it cannot be independently verified from this report (stored explanation, verbatim)
ECR-009	pi·D_R·N vs preferred range and vendor limit	Neither preferred tip-speed range nor vendor limit supplied — never assumed
ECR-009	maxAllowableShaftPower	No allowable shaft power supplied — never assumed
ECR-009	No tip-speed criteria supplied	Neither preferred tip-speed range nor vendor limit supplied — never assumed
ECR-009	maxUnsupportedShaftLength	No shaft-length datum supplied — never assumed

Equation Ref	Item (stored basis string)	Why it cannot be independently verified from this report (stored explanation, verbatim)
ECR-003	Stator open-area fraction	statorOpenAreaFraction not supplied — never assumed
ECR-003	Vendor stator-velocity limits	No stator geometry — check Not Calculable

20. Assumptions Register

Every value below carries source type Assumed or a Thermopac screening default and is Pending Validation. This register is extracted mechanically from the tagged source data — it cannot be edited in the report layer.

Item	Value	Source Reference	Validation Status
Fluid property data contain Assumed entries (RRBO entered tags and/or EPD data)	—	All load- and rotor-derived items are Pending Validation	Assumed — Pending Validation
Continuous-phase viscosity 0.0009170000000000001 Pa·s is ASSUMED	—	Rotor Reynolds number is Pending Validation	Assumed — Pending Validation
Interfacial tension 0.01 N/m is ASSUMED	—	Rotor Weber number is Pending Validation	Assumed — Pending Validation
Rotor speed is ASSUMED	—	All rotor-derived items are Pending Validation	Assumed — Pending Validation
Input tagged ASSUMED (Thermopac Preliminary Equipment Screening Default v1.0, value 1)	—	Downstream items are Pending Validation	Assumed — Pending Validation
Input tagged ASSUMED (Thermopac Preliminary Equipment Screening Default v1.0, value 0.4)	—	Downstream items are Pending Validation	Assumed — Pending Validation
Input tagged ASSUMED (Thermopac Preliminary Equipment Screening Default v1.0, value 0.25)	—	Downstream items are Pending Validation	Assumed — Pending Validation
Input tagged ASSUMED (Thermopac Preliminary Equipment Screening Default v1.0, value 0.9)	—	Downstream items are Pending Validation	Assumed — Pending Validation
Input tagged ASSUMED (Thermopac Preliminary Equipment Screening Default v1.0, value 1.25)	—	Downstream items are Pending Validation	Assumed — Pending Validation
Input tagged ASSUMED (Thermopac Preliminary Equipment Screening Default v1.0, value 1)	—	Downstream items are Pending Validation	Assumed — Pending Validation
Top Head allowance 0.5 m is ASSUMED	—	Height breakdown is Pending Validation	Assumed — Pending Validation
Top Disengagement allowance 1 m is ASSUMED	—	Height breakdown is Pending Validation	Assumed — Pending Validation
Top Distributor allowance 0.5 m is ASSUMED	—	Height breakdown is Pending Validation	Assumed — Pending Validation
Bottom Distributor allowance 0.5 m is ASSUMED	—	Height breakdown is Pending Validation	Assumed — Pending Validation
Bottom Disengagement allowance 1 m is ASSUMED	—	Height breakdown is Pending Validation	Assumed — Pending Validation
Bottom Head allowance 0.5 m is ASSUMED	—	Height breakdown is Pending Validation	Assumed — Pending Validation
Drive/Seal/Bearing allowance 1 m is ASSUMED	—	Height breakdown is Pending Validation	Assumed — Pending Validation

Item	Value	Source Reference	Validation Status
Rotor geometry (diameter/ratio) is ASSUMED	—	All rotor-derived items are Pending Validation	Assumed — Pending Validation
Rotor-to-column diameter ratio	0.5 -	Thermopac Preliminary Equipment Screening Default v1.0	Assumed — Pending Validation
Compartment height	0.25 m	Thermopac Preliminary Equipment Screening Default v1.0	Assumed — Pending Validation
Compartment efficiency	0.4 -	Thermopac Preliminary Equipment Screening Default v1.0	Assumed — Pending Validation
Interfacial tension	0.01 N/m	Two-Phase Properties workspace entry (engineer source type: Assumed)	Assumed — Pending Validation

21. Validation & Missing-Data Summary

Severity	Item	Reason
WARNING	ECR-specific Vendor Hydraulic Capacity	Not supplied — hydraulic utilization cannot be screened at any diameter; the C3 generic percentage is not a substitute (ECR-002).
WARNING	Tip-speed criteria (preferred range / vendor limit)	Neither supplied — tip-speed classification Not Calculable; never assumed (ECR-009).
WARNING	Stator geometry (open-area fraction, velocity limits)	Not supplied — stator free-area velocity and stator-velocity-limit check Not Calculable (ECR-003).
WARNING	Maximum allowable shaft power	Not supplied — shaft-power check Not Calculable (ECR-009).
WARNING	Maximum unsupported shaft length	Not supplied — shaft-support check Not Calculable (ECR-009).
WARNING	Vendor/mechanical bearing data	Bearing/support requirements are Pending Validation — no bearing design is performed (ECR-009).
WARNING	Power number (laboratory/vendor)	N_P = 1 is an Assumed screening default — agitation power scales linearly with it; vendor/laboratory confirmation required (ECR-006).
WARNING	System derating factor	Not supplied — 1.0 applied with warning; never invented.

22. References

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23. Revision History

Report Rev	Date	Generated By	Description
Rev 0	2026-08-06	Manager	Superseded generation
Rev 0	2026-08-06	Manager	Regenerated from current design revision data